Mixed Mahler Congruences

Exact low-precision kernels, improved norm bounds,
and optimal relations for a lacunary sum plus a Cantor product

KEIO*

26 October 2026

Abstract

Let $A(z)=\sum_{n\ge 0} z^{3^n}$ and $B(z)=\prod_{n\ge 0}(1+z^{3^n})$. We study their joint algebra modulo powers of 3 over $R=\mathbb{Z}[z,z^{-1},(1+z)^{-1}]$. The individual minimum-degree problems of Krattenthaler and Müller provide a starting point, but do not determine mixed relations. We give exact joint evaluation kernels modulo $3,9,27,81$, with respectively $6,18,36,48$ necessary and sufficient module generators. For $H=A+B$, the minimum degree of a monic annihilator is exactly 6γ for $1\le\gamma\le 9$, and exactly 54 for $\gamma=10$. The lower bounds use two independent Artin–Schreier correction classes. For every polynomial $F(A,B)$, a root-counting norm construction gives an annihilator of degree at most $6d$, where $d+\sum_{j\ge 1}\lfloor d/9^j\rfloor\ge\gamma$. This is $(16/3)\gamma+O(\log\gamma)$, improving the elementary 6γ bound. We also give an all-precision mixed spanning theorem, classify the reduced mixed fields for every odd prime, and supply a direct characteristic-zero independence proof. The full mixed kernel and the optimal degree for H beyond precision 3^{10} are left open.

1 Introduction and precise scope

Two complementary series used in prime-power congruence methods are

$$A(z)=\sum_{n\ge 0} z^{3^n},\qquad B(z)=\prod_{n\ge 0}(1+z^{3^n}). \tag{1}$$

They satisfy the exact identities

$$A(z^3)=A(z)-z,\qquad B(z^3)=\frac{B(z)}{1+z}. \tag{2}$$

Throughout, congruence of Laurent series means coefficientwise congruence. Our fixed coefficient ring is

$$R=\mathbb{Z}[z,z^{-1},(1+z)^{-1}]\subset\mathbb{Z}((z)).$$

For $H\in\mathbb{Z}((z))$ admitting monic relations modulo every 3^γ, write

$$\mu_H(\gamma)=\min\{\deg_T G : G\in R[T]\text{ monic},\ G(H)\equiv 0 \pmod{3^\gamma}\}. \tag{3}$$

For a Laurent series or an element of R, ν_3 is the least 3-adic valuation of its coefficients; $\nu_3(0)=\infty$. The notation $R[A,B] \bmod 3^\gamma$ always denotes the image under coefficientwise reduction, equivalently $R[X,Y]/J_\gamma$ for the evaluation ideal defined below. It does not denote

*Research draft prepared with AI assistance. The proofs and finite checks are supplied for independent review; the manuscript is not externally peer reviewed or proof-assistant certified.

the intrinsic quotient $R[A,B]/3^\gamma R[A,B]$; coefficientwise divisibility need not be divisibility inside that subalgebra.

Krattenthaler and Müller posed two separate minimum-degree conjectures. Their Conjecture 4 in [1] concerns $\Phi_p(z) = \sum z^{p^n}$ in $\mathbb{Z}[z,T]$: the proposed minimum is the least D with $\nu_p(D!) \geq \gamma$. Their Conjecture 12 in [2] concerns B in the ring R: the proposed minimum is $2d_\gamma$, where

$$w(d) = d + \nu_3(d!), \qquad d_\gamma = \min\{d \geq 1 : w(d) \geq \gamma\}. \tag{4}$$

The arithmetic part of [3] supplies a Frobenius construction attaining both prior lower bounds. Section 2 reproduces the construction and its coefficient-ring descent so that the present mixed results do not depend on an unexamined assertion in that draft.

The substantive joint assertions in this paper are the following.

1. The evaluation ideal in $R[X,Y]$ is computed exactly through precision 81 (Theorem 5.3). The associated counts are minimal numbers of generators, not free ranks.
2. For $H = A + B$,
$$\mu_H(\gamma) = \begin{cases} 6\gamma, & 1 \leq \gamma \leq 9, \\ 54, & \gamma = 10. \end{cases} \tag{5}$$
See Theorem 7.3. This does not follow by multiplying the two individual minimum degrees.
3. Set
$$W_9(d) = \sum_{j \geq 0} \left\lfloor \frac{d}{9^j} \right\rfloor, \qquad e_\gamma = \min\{d \geq 1 : W_9(d) \geq \gamma\}. \tag{6}$$
For every $F \in R[X,Y]$, the series $F(A,B)$ has a monic annihilator of degree $6e_\gamma$ modulo 3^γ (Theorem 6.2). No affine functional equation for $F(A,B)$ is assumed.

We separate these joint conclusions from standard consequences of Galois theory and Mahler's method, which are included to explain the structure. We do not identify or claim the resolution of an additional previously published conjecture by the mixed results. In particular, the Apéry congruence conjectures at the end of [2] were already settled in [4]. The questions in Section 9 are questions posed here, not relabelled historical conjectures.

2 Individual Frobenius constructions and descent

2.1 The lifting identity

Let p be a prime and let a coefficient ring of integral Laurent series be stable under $\sigma : c(z) \mapsto c(z^p)$. Suppose $f(z^p) = a(z)f(z) + b(z)$, and define

$$\tau G(z,T) = G(z^p, a(z)T + b(z)).$$

Lemma 2.1 (Frobenius correction)**.** *If G is monic of degree $D > 0$ and $G(z, f(z))$ is divisible by p^e, then*

$$G^p - p^{(p-1)e}\tau G$$

is monic of degree pD and its evaluation is divisible by p^{pe+1}. Starting with degree D_0 and divisibility p^{e_0} gives degrees $D_0 p^r$ and divisibilities p^{e_r}, where $e_r = p^r e_0 + (p^r - 1)/(p-1)$.

Proof. Write $G(f) = p^e C(z)$. Evaluation commutes with the substitutions, so

$$(G^p - p^{(p-1)e}\tau G)(f) = p^{pe}(C(z)^p - C(z^p)).$$

The parenthesis is divisible by p by the coefficientwise Frobenius congruence. The correction has degree at most $D < pD$, which preserves monicity. The recurrence $e_{r+1} = pe_r + 1$ gives the formula. □

If $d = \sum d_r p^r$ is a base-p expansion, multiplying the corresponding tower polynomials to powers d_r gives degree $D_0 d$ and valuation at least

$$\sum_r d_r e_r = e_0 d + \nu_p(d!). \tag{7}$$

The digit-product step was already described in the original papers; it is the recursively available tower that supplies all the inputs.

For Φ_p, take the seed T, $e_0 = 0$, and $a = 1, b = -z$. This gives monic polynomials of every degree i, with evaluation valuation at least $\nu_p(i!)$. Combined with the lower bound in [1, Lemma 1], this attains the value in its Conjecture 4. The construction applies to every prime, not only 3.

2.2 The Cantor ring and its finite descent

For B, initially work in

$$R_\infty = \mathbb{Z}[z, z^{-1}, (1 + z^{3^j})^{-1} : j \geq 0].$$

This ring is stable under $z \mapsto z^3$. Frobenius and (2) imply $B^2 \equiv (1+z)^{-1} \pmod 3$. Start with $T^2 - (1+z)^{-1}$, degree 2 and valuation at least 1. Lemma 2.1 and (7) give a monic polynomial of degree $2d$ with evaluation valuation at least $w(d)$.

We must return to the prescribed ring R, rather than silently enlarging it. Put $U_j = 1 + z^{3^j}$, $V_j = (1+z)^{3^j}$, and $D_j = (U_j - V_j)/3 \in \mathbb{Z}[z]$. For every $N \geq 1$,

$$U_j^{-1} \equiv \sum_{a=0}^{N-1} (-3D_j)^a V_j^{-a-1} \pmod{3^N}. \tag{8}$$

Indeed the exact remainder is $(-3D_j)^N V_j^{-N} U_j^{-1}$. Substitution in each of the finitely many coefficients descends a polynomial from R_∞ to R to any fixed precision. Its leading coefficient 1 is retained unchanged. The lower bound of [2, Theorem 10] now gives the asserted minimum $2d_\gamma$ in its Conjecture 12.

Proposition 2.2 (Two triangular families)**.** *There are fixed monic polynomials $a_i \in R[X]$ and $b_j \in R[Y]$, of degrees i, j, such that*

$$\nu_3(a_i(A)) \geq \alpha_i := \nu_3(i!), \qquad \nu_3(b_j(B)) \geq \beta_j := w(\lfloor j/2 \rfloor).$$

They may be chosen algorithmically by the preceding towers and finite descent, using precision $\beta_j + 1$ for each b_j.

Proof. The first construction was just given. For $j = 2d$, use the descended Cantor polynomial; for $j = 2d+1$, multiply it by Y. Since B is a unit, the valuation lower bound is unchanged. □

Only these lower bounds, not an unproved tensor-product independence assertion, are used in the next section. Also

$$\alpha_{3r+s} = w(r) \quad (0 \leq s < 3), \qquad \beta_{2j+\epsilon} = w(j) \quad (0 \leq \epsilon < 2). \tag{9}$$

3 Mixed spanning at arbitrary precision

Theorem 3.1 (Weighted mixed spanning)**.** *Modulo 3^γ, every polynomial in A, B with coefficients in R is an R-linear combination of*

$$a_i(A)b_j(B), \qquad \alpha_i + \beta_j < \gamma.$$

Consequently it is generated by at most

$$N_\gamma = 6\#\{(r,j) \in \mathbb{Z}^2_{\geq 0} : w(r) + w(j) < \gamma\} \tag{10}$$

elements. As $\gamma \to \infty$,

$$N_\gamma = \frac{4}{3}\gamma^2 + O(\gamma \log \gamma).$$

Proof. Monicity makes the products $a_i(X)b_j(Y)$ an R-basis of $R[X,Y]$ by triangular change of basis from the monomials. Terms whose weight is at least γ vanish upon evaluation modulo 3^γ. Equation (9) gives the count. Legendre's formula gives $w(r) = (3r - s_3(r))/2 = (3/2)r + O(\log(r+2))$. The permitted pairs therefore differ by only an $O(\log \gamma)$-wide boundary strip from the integer points in $r + j < 2\gamma/3$. The strip contains $O(\gamma \log \gamma)$ points; the triangle contains $2\gamma^2/9 + O(\gamma)$ points. Multiplication by 6 proves the formula. □

Separate monic reductions instead give a rectangle of $(3d_\gamma)(2d_\gamma) = 6d_\gamma^2$ terms, asymptotic to $(8/3)\gamma^2$. Some sample counts are:

γ	d_γ	Rectangle	N_γ	6γ
1	1	6	6	6
2	2	24	18	12
3	3	54	36	18
4	3	54	48	24
10	8	384	216	60
100	69	28566	14646	600

The last column is an elementary *annihilator-degree* bound for one mixed element, not a count of simultaneous module generators. These are different optimization problems. In fact, a monic relation of degree 6 exists modulo 3 for every mixed element, and its γ-th power gives the last column. Section 6 improves that bound. We assert sharpness of N_γ only for $\gamma \leq 4$.

4 The reduced mixed field

The following description also clarifies why separate reductions do not settle the mixed problem. We state it for every odd prime p. Let x, y be the coefficientwise reductions of $\sum z^{p^n}$ and $\prod(1 + z^{p^n})$, and put $K = \mathbb{F}_p(z)$. They satisfy

$$x^p - x = -z, \qquad y^{p-1} = (1+z)^{-1}. \tag{11}$$

Theorem 4.1 (Mixed degree classification)**.** *The field $L = K(x,y)$ is Galois over K, of degree $p(p-1)$, with automorphisms $x \mapsto x + c$, $y \mapsto \zeta y$ for $c \in \mathbb{F}_p$, $\zeta \in \mathbb{F}_p^\times$. For a nonzero $f \in L$, write uniquely*

$$f = \sum_{j=0}^{p-2} f_j(x) y^j, \qquad f_j \in K[X], \quad \deg f_j < p.$$

Set $d = \gcd(p-1, \{j : f_j \neq 0\})$, $m = (p-1)/d$, *and let* $\epsilon = 0$ *if every* f_j *is constant in* X, *and* $\epsilon = 1$ *otherwise. Then*

$$[K(f) : K] = p^{\epsilon} m, \qquad K(f) = \begin{cases} K(y^d), & \epsilon = 0, \\ K(x, y^d), & \epsilon = 1. \end{cases}$$

The genus of this field is 0 *if* $\epsilon = 0$, *and* $(p-1)(m-1)/2$ *if* $\epsilon = 1$.

Proof. The Artin–Schreier polynomial for x is irreducible: a rational function $u^p - u$ cannot have a pole of order 1 at infinity. Here the absence of a rational root suffices: after adjoining one root all the roots are obtained by adding elements of $\mathbb{F}_p$, so its splitting-field Galois group is a subgroup of the prime-order group $(\mathbb{F}_p, +)$. The polynomial for y^{-1} is Eisenstein at $z+1$, of degree $p-1$. The two Galois degrees are coprime, so their compositum has the asserted group.

Translation invariance of a polynomial in x of degree less than p forces it to be constant, as its highest nonconstant term has nonzero finite difference. The Kummer subgroup fixing f has exactly d elements. Every subgroup of the direct product of these coprime cyclic groups splits into its two components. Thus the stabilizer has order $p^{1-\epsilon}d$, proving the degree and the stated fixed field.

If $\epsilon = 0$, then $z = (y^d)^{-m} - 1$, so the field is rational. Otherwise set $U = x$ and $V = y^{-d}$; its equation is $V^m = 1 + U - U^p$. The right side has p simple roots and a pole of order p at infinity. The tame Kummer cover is fully ramified at these $p+1$ points. Riemann–Hurwitz gives $2g - 2 = -2m + (p+1)(m-1)$, as asserted. □

Corollary 4.2 (Sum and product)**.** *The following are irreducible minimal polynomials over* K*:*

$$\mu_{x+y}(T) = (T^p - T + z)^{p-1} - \frac{z^{p-1}}{(1+z)^p},$$
$$\mu_{xy}(T) = (T^p - qT)^{p-1} - z^{p-1}q^p, \qquad q = (1+z)^{-1}.$$

Both have degree $p(p-1)$.

Proof. For the sum, $(x+y)^p - (x+y) + z = -zy/(1+z)$. For the product, $(xy)^p = qxy - zqy$. Taking $(p-1)$-st powers gives the displayed relations. The degree classification, or the distinct conjugates, proves minimality. □

For $p = 3$, put $h = x + y$ and retain $K = \mathbb{F}_3(z)$, $L = K(x, y)$. Then $[L : K] = 6$, $L = K(h)$, and

$$M(T) = (T^3 - T + z)^2 - \frac{z^2}{(1+z)^3} \in R[T] \tag{12}$$

reduces to the minimal polynomial of h. This degree-6 field has genus 1; its proper intermediate fields have genus 0.

5 Independent first corrections and exact mixed kernels

Define

$$P(X) = X^3 - X + z, \qquad E(Y) = (1+z)Y^2 - 1, \tag{13}$$

and define integral power series and their reductions by

$$C_0 = \frac{P(A)}{3}, \quad W_0 = \frac{E(B)}{3}, \qquad c = \overline{C_0}, \quad w = \overline{W_0}.$$

Lowercase w here is a series; the weight function $w(d)$ always has an integer argument.

Lemma 5.1 (Two independent correction classes)**.** *Over L, the elements c, w satisfy*

$$c^3 - c = -zx^2 + z^2x = x^7 - x^5 =: f, \qquad w^3 - w = -\frac{z}{(1+z)^2} =: g. \tag{14}$$

Their compositum has degree 9 over L, with Galois group $\mathbb{F}_3^2$ acting by independent translations of c, w.

Proof. Direct substitution in (2) gives

$$C_0(z^3) = C_0(z) - zA(A - z),$$

$$W_0(z^3) = W_0(z) - \frac{z}{(1+z)^2} - \frac{3z}{(1+z)^2}W_0(z).$$

Reduction modulo 3 gives (14). Since $z = x - x^3$, its first right side equals $x^7 - x^5$.

It remains to check independence of the two Artin–Schreier classes. We may extend constants to an algebraic closure for this check. The field L has equation $y^{-2} = 1 + x - x^3$. At the place over $x = \infty$, x has pole order 2, hence f has pole order 14; g is regular there. At any place over $1 + z = 0$, the quadratic Kummer extension has ramification index 2, so g has pole order 4; f is regular there. Both 14 and 4 are prime to 3. A function of the form $u^3 - u$ has pole order divisible by 3 at every pole. Thus no nonzero $\mathbb{F}_3$-linear combination of f, g is of that form: use infinity if the coefficient of f is nonzero, and a place over -1 otherwise. The two classes are independent. Equivalently, their two cyclic degree-3 extensions are distinct and linearly disjoint, giving the asserted degree and group. □

Lemma 5.2 (Exact valuation in a finite box)**.** *For arbitrary $d_{s,e,r,j} \in R$, set*

$$D(X, Y) = \sum_{\substack{0 \le s,r,j \le 2 \\ 0 \le e \le 1}} d_{s,e,r,j} X^s Y^e P(X)^r E(Y)^j.$$

Then

$$\nu_3(D(A, B)) = \min_{s,e,r,j} \big(\nu_3(d_{s,e,r,j}) + r + j\big). \tag{15}$$

In particular the formula is exact for every polynomial with $\deg_X < 9$, $\deg_Y < 6$, written in this triangular basis.

Proof. The zero polynomial is immediate. Otherwise divide the evaluation by the power of 3 given by the minimum, and reduce modulo 3. The terms attaining the minimum become a linear combination of $x^s y^e c^r w^j$. The six elements $x^s y^e$ form a K-basis of L, and the nine elements $c^r w^j$ form an L-basis by Lemma 5.1. Their 54 products are K-independent, so the reduced combination is nonzero. Division of elements of R by their exact powers of 3 stays in R, since its inverted denominators have nonzero reductions. This proves the equality. □

Put $\mathfrak{m} = (3, P, E) \subset R[X, Y]$ and

$$U(X) = P(X)^3 - 9P(X) + 27zX(X - z), \tag{16}$$

$$V(Y) = E(Y)^3 - 9E(Y) + \frac{27z}{(1+z)^2}\big(E(Y) + 1\big). \tag{17}$$

Their evaluations are divisible by 81, by (14) or the exact functional identities in its proof. The polynomial U is monic of degree 9 in X; V has degree 6 in Y and leading coefficient $(1 + z)^3$, a unit in R.

Theorem 5.3 (Exact kernels through 81)**.** *Let*

$$J_\gamma = \{G \in R[X,Y] : G(A,B) \equiv 0 \pmod{3^\gamma}\}.$$

Then

$$J_1 = \mathfrak{m}, \qquad J_2 = \mathfrak{m}^2, \qquad J_3 = \mathfrak{m}^3, \qquad J_4 = \mathfrak{m}^4 + (U,V). \tag{18}$$

For $1 \le \gamma \le 4$, the evaluated algebra has the following decomposition as an R-module:

$$R[A,B] \bmod 3^\gamma \;\cong \bigoplus_{\substack{0\le r,j\le 2\\ r+j<\gamma}} \left(R/3^{\gamma-r-j}R\right)^6. \tag{19}$$

Its minimum number of R-module generators is respectively $6, 18, 36, 48$.

Proof. The displayed generators of the ideals have the required evaluation divisibility, which proves one inclusion. Divide an arbitrary polynomial by U and by $(1+z)^{-3}V$ to obtain a remainder of degrees $< 9, < 6$. These divisions commute since each divisor uses only its own polynomial variable. Both discarded terms evaluate to multiples of 81. Expand the remainder as in Lemma 5.2. Its evaluation vanishes modulo 3^γ precisely when each coefficient is divisible by $3^{\max(\gamma-r-j,0)}$. Every resulting term belongs to $\mathfrak{m}^\gamma$. Also $U, V \in \mathfrak{m}^3$, so for $\gamma \le 3$ the discarded terms already belong to $\mathfrak{m}^\gamma$; for $\gamma = 4$ they are the two additional generators. This proves (18).

The same exact coefficient criterion proves the module decomposition. After quotienting this module by 3 and extending scalars from $R/3R$ to K, its dimension is the number of summands. Any generating set must have at least that many elements, and the displayed basis classes attain the count. These are cyclic torsion summands, not a claim that the algebra is a free module over $R/3^\gamma R$. □

6 A norm construction at every precision

We first prove a root-counting lemma that is useful beyond the present pair of series. The valuation argument takes place in a complete discretely valued field $\mathcal{K}$ of characteristic 0, with valuation ring $\mathcal{O}$, uniformizer 3, and an isometric endomorphism σ inducing Frobenius on the residue field. For our series one can take the fraction field of

$$\mathcal{O} = \varprojlim_n (\mathbb{Z}/3^n\mathbb{Z})((z)),$$

with $\sigma(z) = z^3$. Thus A, B, R_∞ embed in $\mathcal{O}$, and their original coefficientwise valuation is unchanged.

Lemma 6.1 (Root counts along a Frobenius tower)**.** *Suppose $a \in \mathcal{O}^\times$, $b, F \in \mathcal{O}$, and $\sigma F = aF + b$. Let $G_0 \in \mathcal{O}[T]$ be monic, with $v(G_0(F)) \ge 1$ and $G_0'(F) \in \mathcal{O}^\times$. Put*

$$e_r = \frac{3^{r+1}-1}{2}, \qquad G_{r+1} = G_r^3 - 3^{2e_r}\tau G_r, \quad \tau G_r(T) = (\sigma G_r)(aT+b).$$

For every $1 \le t \le r+1$, exactly 3^{r-t+1} roots α of G_r, counted with multiplicity, satisfy $v(\alpha - F) \ge t$. All roots are integral.

Proof. Monicity and integrality give the last assertion. Define

$$v_{r,t} = \sum_{j=1}^{t} 3^{r-j+1}, \qquad Z_{r,t}(Y) = 3^{-v_{r,t}} G_r(F + 3^t Y).$$

We prove that $Z_{r,t}$ is integral and its reduction has degree 3^{r-t+1}, and that for $t = r+1$ this reduction is affine with unit linear coefficient. For $r = 0, t = 1$, this follows by Taylor expansion and the two hypotheses on G_0.

For $t \le r$, the recurrence gives

$$Z_{r+1,t} = Z_{r,t}^3 - 3^{2(e_r - v_{r,t})}(\sigma Z_{r,t})(aY).$$

Here $e_r - v_{r,t} \ge 1$ and $v_{r+1,t} = 3v_{r,t}$, so its reduction is the cube of the previous reduction, of the required degree. At $t = r+1$, we have $v_{r,t} = e_r$. Write $\overline{Z}_{r,r+1} = h + uY$, with $u \neq 0$. Scaling the next polynomial by 3^{3e_r} gives reduction

$$(h+uY)^3 - (h^3 + u^3\bar{a}Y) = u^3Y^3 - u^3\bar{a}Y.$$

It has degree 3 and unit linear coefficient. Substitute $Y \mapsto 3Y$ and divide by 3 for the next depth. Its constant coefficient is integral: putting $C = 3^{-e_r}G_r(F) \in \mathcal{O}$ gives $G_{r+1}(F) = 3^{3e_r}(C^3 - \sigma C) \in 3^{3e_r+1}\mathcal{O}$. Its linear coefficient is a unit and the higher coefficients vanish on reduction. This proves the induction.

For completeness, a primitive integral polynomial has exactly as many integral roots as the degree of its reduction, counted with multiplicity. In a finite valued splitting extension, factor it as

$$c \prod_{v(\alpha)\ge 0} (Y - \alpha) \prod_{v(\alpha)<0} (1 - Y/\alpha),$$

absorbing the factors $-\alpha$ into c. Every displayed factor is integral and primitive. Their reductions have degrees 1 and 0, respectively. The product is primitive because its reduction is nonzero, so c is a unit. Its reduced degree therefore counts exactly the integral roots. Apply it to $Z_{r,t}$. Integral values of $Y = (\alpha - F)/3^t$ are exactly the roots in the asserted ball. □

For the pair A, B, take the two seed polynomials

$$G_0(X) = X^3 - X + z, \qquad Q_0(Y) = Y^2 - (1+z)^{-1},$$

and the substitutions $X \mapsto X - z$, $Y \mapsto Y/(1+z)$. Both derivatives at the corresponding series are units modulo 3. Use Lemma 6.1 to form G_r, Q_r; they are monic of degrees $3 \cdot 3^r$ and $2 \cdot 3^r$ over R_∞.

Theorem 6.2 (Mixed norm bound)**.** *For every $F \in R[X,Y]$ and $r \ge 0$ there is a monic polynomial $N_{r,F} \in R[T]$ of degree $6 \cdot 9^r$ such that*

$$\nu_3\big(N_{r,F}(F(A,B))\big) \ge \frac{9^{r+1} - 1}{8}. \tag{20}$$

Consequently

$$\mu_{F(A,B)}(\gamma) \le 6e_\gamma = \frac{16}{3}\gamma + O(\log \gamma), \tag{21}$$

where e_γ is defined in (6).

Proof. The quotient

$$\mathcal{A}_r = R_\infty[X,Y]/(G_r(X), Q_r(Y))$$

is free of rank $6 \cdot 9^r$. Take the characteristic polynomial $\widetilde{N}_{r,F}(T)$ of multiplication by F on this module. It is monic with coefficients in R_∞. In a splitting field it is

$$\prod_{\alpha: G_r(\alpha)=0} \prod_{\beta: Q_r(\beta)=0} (T - F(\alpha,\beta)),$$

with multiplicities. Polynomial evaluation with integral coefficients is valuation-Lipschitz:

$$v(F(A,B)-F(\alpha,\beta)) \geq \min\{v(A-\alpha), v(B-\beta)\}.$$

For each integer depth $t = 1, \ldots, r+1$, the root-counting lemma gives 9^{r-t+1} pairs with both distances at least t. All other factors have nonnegative valuation. Summing these contributions gives

$$v(\widetilde{N}_{r,F}(F(A,B))) \geq \sum_{t=1}^{r+1} 9^{r-t+1} = \frac{9^{r+1}-1}{8}.$$

Descend the finitely many coefficients by (8) to this precision, preserving monicity and degree, to obtain $N_{r,F} \in R[T]$.

For $d = \sum d_r 9^r$, multiply $N_{r,F}^{d_r}$. Its degree is $6d$ and its valuation is at least

$$\sum_r d_r \frac{9^{r+1}-1}{8} = \sum_{j\geq 0} \left\lfloor \frac{d}{9^j} \right\rfloor = W_9(d).$$

Choose $d = e_\gamma$. Finally $W_9(d) = (9d - s_9(d))/8$ implies $e_\gamma = (8/9)\gamma + O(\log \gamma)$. □

Remark 6.3. This is a univariate annihilator bound for each fixed mixed element. It improves 6γ beginning at $\gamma = 10$: the first two tower points are degree 54 at precision 3^{10} and degree 486 at precision 3^{91}. It is not a proof that $6e_\gamma$ is minimal at arbitrary precision, nor a complexity bound for expanding the resultant or determinant.

7 Optimal relations for the sum through 3^{10}

Let $H = A + B$ and let M be the monic degree-6 polynomial in (12). Its evaluation is divisible by 3; put $Z_0 = M(H)/3$ and $v = \overline{Z_0}$. The next lemma detects the first genuinely mixed obstruction.

Lemma 7.1 (The sum correction is primitive)**.** *Let $a = (1+z)^{-1}$ and $h = x + y$. Then*

$$v = zay\,c + za^2 w + za^2 xh. \tag{22}$$

Moreover $[L(v) : L] = 9$ and $[K(h, v) : K] = 54$.

Proof. In characteristic zero use $B^2 = a(1 + 3W_0)$ and set

$$J = C_0 + aBW_0 + ABH.$$

Then $H^3 - H + z = -zaB + 3J$, whence the exact identity

$$\frac{M(H)}{3} = z^2 a^3 W_0 - 2zaBJ + 3J^2.$$

On reducing modulo 3, use $y^2 = a$ and $-2 = 1$ to obtain (22). An automorphism $(i, j) \in \mathbb{F}_3^2$ of $L(c, w)/L$ translates v by $zay\,i + za^2 j$. The ratio of these two coefficients is $y/a = (1 + z)y$, whose square is $1 + z$. It is not in $\mathbb{F}_3$. Therefore no nonzero pair (i, j) fixes v, so its orbit has size 9. Finally $L = K(h)$ has degree 6. □

Proposition 7.2 (Exact valuation below degree 54)**.** *Every $G \in R[T]$ of degree less than 54 has a unique expansion*

$$G(T) = \sum_{r=0}^{8} g_r(T) M(T)^r, \qquad \deg g_r < 6.$$

Writing $\nu_3(g_r)$ for the minimum valuation of its coefficients,

$$\nu_3(G(H)) = \min_{0\le r\le 8}\{\nu_3(g_r) + r\}. \tag{23}$$

Proof. The expansion is monic polynomial division. Divide the evaluation by the least indicated power of 3 and reduce. The resulting combination is in the 54 elements $h^s v^r$, $0 \le s < 6$, $0 \le r < 9$. They are independent over K by Lemma 7.1, so at least one surviving coefficient prevents cancellation. □

Theorem 7.3 (The exact initial degree sequence)**.** *For $H = A + B$, formula (5) holds in the original ring R.*

Proof. For $1 \le \gamma \le 9$, the polynomial M^γ is a monic annihilator of degree 6γ. If a monic G had degree $D < 6\gamma \le 54$, its highest nonzero M-coefficient would be monic and have index $\lfloor D/6 \rfloor \le \gamma - 1$. Proposition 7.2 would give $\nu_3(G(H)) \le \gamma - 1$, a contradiction.

For $\gamma = 10$, Theorem 6.2 with $r = 1$ provides a monic degree-54 annihilator. Every monic polynomial of degree less than 54 has valuation at most 8 by the same argument, so no smaller degree is possible. □

An explicit compact prescription for the degree-54 polynomial is

$$\begin{aligned} G_1(X) &= (X^3 - X + z)^3 - 9((X - z)^3 - (X - z) + z^3), \\ Q_1(Y) &= \left(Y^2 - \frac{1}{1+z}\right)^3 - 9\left(\frac{Y^2}{(1+z)^2} - \frac{1}{1+z^3}\right), \\ \widetilde{N}(T) &= \mathrm{Res}_X\big(G_1(X), Q_1(T - X)\big). \end{aligned}$$

This is monic of degree 54. Descent of $(1 + z^3)^{-1}$ by (8) with $N = 10$, before or after taking the resultant, places its coefficients in R without changing its evaluation modulo 3^{10}. The construction does not require printing a large expanded polynomial.

γ	1	2	3	4	5	6	7	8	9	10
$\mu_H(\gamma)$	6	12	18	24	30	36	42	48	54	54

The different first improvement thresholds are informative: individual relations improve on their naive powers at $\gamma = 4$, while the mixed sum does not improve on M^γ until $\gamma = 10$.

8 Characteristic-zero independence and specialization

The arithmetic dependence modulo prime powers should not be confused with dependence in characteristic zero. The next argument is a direct specialization of established asymptotic ideas in Mahler theory. The underlying independence also falls under classical mixed additive–multiplicative criteria; see [5, 6].

Theorem 8.1 (A direct radial proof)**.** *For an integer $b \ge 2$ that is not a power of 2, the functions*

$$A_b(z) = \sum_{n\ge 0} z^{b^n}, \qquad B_b(z) = \prod_{n\ge 0}(1 + z^{b^n})$$

are algebraically independent over $\mathbb{C}(z)$.

Proof. Set $z = e^{-t}$, $t \downarrow 0$, and $N = \lfloor \log_b(1/t) \rfloor$. Summing $1 - e^{-b^n t} = O(b^n t)$ for $n < N$ and the exponentially decaying tail for $n \geq N$ gives

$$A_b(e^{-t}) = \frac{\log(1/t)}{\log b} + O(1).$$

Similarly $\log(1 + e^{-b^n t}) = \log 2 + O(b^n t)$ for $n < N$, and the remaining logarithms have a uniformly bounded sum. Thus, with $\theta = \log_b 2$, there are positive constants c, C such that

$$ct^{-\theta} \leq B_b(e^{-t}) \leq Ct^{-\theta}.$$

Suppose a nonzero polynomial relation exists, and clear denominators:

$$\sum_{j=0}^{J} q_j(z, A_b(z)) B_b(z)^j = 0, \qquad q_j \in \mathbb{C}[z, X].$$

For each nonzero $q_j = \sum_i q_{ij}(z) X^i$, let $m_j = \min_i \operatorname{ord}_{z=1} q_{ij}$ and choose the largest $i = d_j$ attaining that minimum. Taylor expansion at 1 and the first asymptotic estimate give

$$q_j(e^{-t}, A_b(e^{-t})) = c_j t^{m_j} (\log(1/t))^{d_j} (1 + o(1)), \quad c_j \neq 0.$$

Because θ is irrational, the exponents $m_j - j\theta$ are pairwise distinct. The term with the smallest exponent dominates all the others in absolute value, even allowing the bounded multiplicative oscillation of B_b. Dividing the relation by that nonzero term gives $1 + o(1) = 0$, a contradiction. Finally $\log_b 2$ is rational for an integer $b \geq 2$ exactly when b is a power of 2. □

Corollary 8.2 (Values at regular algebraic points)**.** *For b as above and every algebraic α with $0 < |\alpha| < 1$, $A_b(\alpha)$ and $B_b(\alpha)$ are algebraically independent over $\overline{\mathbb{Q}}$. In particular, for $b = 3$ their sum, product, and every nonconstant polynomial in the two values with algebraic coefficients are transcendental.*

Proof. Apply Nishioka's specialization theorem, in the form of [7, Theorem 1], to

$$\begin{pmatrix} 1 \\ A_b(z) \\ B_b(z) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ z & 1 & 0 \\ 0 & 0 & 1+z \end{pmatrix} \begin{pmatrix} 1 \\ A_b(z^b) \\ B_b(z^b) \end{pmatrix}.$$

The matrix and its inverse are regular at every iterate α^{b^n}, since these points are nonzero and have modulus less than 1. Its determinant is $1 + z$. The functional transcendence degree is 2 by Theorem 8.1, so the same is true for the values. A nonconstant algebraic-coefficient polynomial value could not be algebraic without providing a nonzero polynomial relation. □

This corollary is an application of a prior theorem, not an additional claimed solution of an open transcendence conjecture. It concerns formal-series values inside the unit disk, and has no implication for the Mills-type constants considered elsewhere in the corpus.

9 What remains open and what has been established

The proof of the low-precision kernel uses two independent first corrections. Higher corrections need further independence arguments; neither the two individual conjectures nor characteristic-zero independence supplies them automatically.

Question 9.1 (Higher mixed evaluation kernels)**.** Determine J_γ for $\gamma \geq 5$. Is there a compatible choice of the triangular families in Proposition 2.2 for which every finite expansion satisfies

$$\nu_3\left(\sum_{i,j} c_{ij}a_i(A)b_j(B)\right) = \min_{i,j}(\nu_3(c_{ij}) + \alpha_i + \beta_j)?$$

The equality, if proved with the coefficient-ring details retained, would make the all-precision spanning count N_γ sharp. The inequality in the direction $\geq$ is already proved here. We do not assume the reverse inequality outside the finite box of Lemma 5.2.

Question 9.2 (The optimal degree sequence of the sum)**.** Does $\mu_{A+B}(\gamma) = 6e_\gamma$ hold for every $\gamma \geq 1$?

Theorem 6.2 proves the upper bound at every precision and Theorem 7.3 proves equality for $\gamma \leq 10$. For $\gamma > 10$ this paper provides no matching general lower bound. An equality assertion in this range would require new work, not an extrapolation from the displayed table.

Question 9.3 (Other primitive mixed elements)**.** For which $F \in R[X, Y]$ is the norm bound optimal? How does the degree of the first normalized correction over $K(\bar{F})$ determine the initial segment of $\mu_{F(A,B)}$?

The sum calculation gives one complete example: the reduced field has degree 6 and the first correction has degree 9 over it. The universal norm construction explains why base 9, rather than base 3, appears in the resulting mixed upper bound.

For attribution, the one-series lower bounds belong to Krattenthaler–Müller, the Frobenius construction and descent were developed in [3], and the Galois, Newton-polygon, and Mahler specialization tools are classical. The contributions developed here are the explicit mixed application of those tools, including the exact kernels, the degree-54 obstruction and construction, and the all-precision mixed norm bound. A scoped literature check did not locate identical joint statements; this is not an assertion of an exhaustive priority search or external acceptance.

A Exact checks and reproducibility

The proofs are infinite-series and field-theoretic arguments. Finite checks are supplementary and cannot establish a minimum-degree lower bound for arbitrary coefficient functions.

Compiling this single TeX source also writes `mixed_mahler_verify.py`, an optional verification script requiring Python 3 and SymPy. Run it with `python3 mixed_mahler_verify.py`. The script checks the symbolic identities below, the first 512 coefficients of the two correction equations and the sum-correction identity, the 81-divisibility of $U(A), V(B)$ at that truncation, and the integer counting tables. It also checks the degree-54 resultant modulo 3^{10} at the convergent 3-adic evaluation points $z = 3, 6, 9, 12$. These checks passed for the present draft; the checks at particular values do not prove the formal-series assertion.

The following identities are convenient independent checks of signs and denominators. All substitutions are simultaneous:

$$P(z^3, X - z) = P(z, X) - 3zX(X - z),$$
$$E\left(z^3, \frac{Y}{1+z}\right) = E(z, Y) - \frac{3z}{(1+z)^2}(E(z, Y) + 1),$$
$$z = x - x^3 \implies -zx^2 + z^2x = x^7 - x^5.$$

They can be verified by polynomial expansion over $\mathbb{Q}(z)$. The first two also verify the displayed formulas for U, V and the Artin–Schreier equations. The exact identity preceding (22) verifies the primitive-correction computation.

For the numerical tables, the following complete Python code uses only integer arithmetic and reproduces the counting formulas. It does not test or replace the independence proofs.

```
def v3fact(n):
    s = 0
    while n:
        n //= 3
        s += n
    return s
def weight(n):
    return n + v3fact(n)
def w9(n):
    s = 0
    while n:
        s += n
        n //= 9
    return s
for g in (1, 2, 3, 4, 10, 100):
    d = next(n for n in range(1, g+1) if weight(n) >= g)
    N = 6 * sum(weight(i)+weight(j) < g
        for i in range(d) for j in range(d))
    e = next(n for n in range(1, g+1) if w9(n) >= g)
    print(g, d, 6*d*d, N, 6*g, 6*e)
```

The last two columns compare the elementary and norm degree upper bounds. At $\gamma = 100$ they are 600 and 540; this is an upper-bound comparison, not a computed exact minimum.

References

[1] C. Krattenthaler and T. W. Müller, *A method for determining the mod-p^k behaviour of recursive sequences*, Aequationes Math. **99** (2025), 2505–2547. doi:10.1007/s00010-025-01195-0. Preprint first posted 2015; revised version: arXiv:1508.02580v2.

[2] C. Krattenthaler and T. W. Müller, *A method for determining the mod-3^k behaviour of recursive sequences*, preprint (2013), arXiv:1308.2856. Theorem 10 and Conjecture 12 refer to this version.

[3] KEIO, *Frobenius Lifting, Floquet Rigidity, and Equivariant Stabilization*, research manuscript [070], Verifiable Mathematical Reasoning Corpus, 24 October 2026, arithmetic sections T76–T78. Public corpus. Source inspected at commit `dd9c67cb72c5991cce5816de540db8c2132ca343`. Unreviewed research draft.

[4] C. Krattenthaler and T. W. Müller, *Generalised Apéry numbers modulo* 9, J. Number Theory **147** (2015), 708–720. arXiv:1401.1444.

[5] K. K. Kubota, *On the algebraic independence of holomorphic solutions of certain functional equations and their values*, Math. Ann. **227** (1977), 9–50.

[6] K. Väänänen, *Algebraic independence of certain Mahler numbers*, preprint (2015), arXiv:1507.02510. See Theorem K for the mixed additive–multiplicative criterion.

[7] B. Adamczewski and C. Faverjon, *A new proof of Nishioka's theorem in Mahler's method*, C. R. Math. **361** (2023), 1011–1028. doi:10.5802/crmath.458.